How much solar can India's cropland carry while keeping 80% of the crop yield?

Light model
3 array layouts → Crop model
DSSAT, 4 crops → Yield rule
≥ 80% of yield

2,977 grid cells · 5 climate models · today and 2050

39.2 TWp on 39.7 Mha of cropland (robust),
≈ 42× India's yearly electricity use

52% | 29% max panel cover,
cereals | legumes

−1% capacity by 2050 (SSP5-8.5),
annual energy −5%

